

MUHAMMAD USMAN

Full-Stack AI Engineer | AI Systems & Autonomous Agents

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SUMMARY

Full-Stack AI Engineer specializing in LLM-powered applications, autonomous agent systems, and high-performance backend tooling. Built and shipped production systems spanning a PyPI-published Rust database engine, RAG-powered codebase intelligence, and multi-channel AI agent platforms — from low-level systems programming to serverless React frontends.

KEY SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Rust, SQL, FastAPI, Next.js, React, Node.js
AI/ML: RAG Pipelines, Agent Architectures, LangChain, Vector Databases, OpenAI, Gemini, Claude
Databases & Tools: Qdrant, PostgreSQL, SQLite, Supabase, Docker, Git, GitHub Actions, Vercel

WORK EXPERIENCE

AI/ML Engineering Intern | DeveloperHub.co

Dec 2025 – Jan 2026

- Shipped 5 production-grade ML/AI Streamlit apps in 6 weeks spanning RAG chatbots and predictive models using Python, LangChain, TensorFlow, and Scikit-learn.
- Architected a Gemini-powered mental health assistant with real-time crisis-detection logic.

PROJECTS

Autonoma | Python, React 19, SQLite FTS5, asyncio, WebSockets

github.com/MuhammadUsmanGM/autonoma

- Built a production-grade AI agent platform cutting manual workload by 85% with a live telemetry HUD and unified gateway across Telegram, WhatsApp, Discord, and Gmail.
- Engineered a BM25-ranked SQLite FTS5 memory engine with importance-based decay and deduplication; shipped as a global npm package achieving sub-5ms multi-session retrieval.

CodeLens | Next.js, Gemini Pro, Qdrant, LangChain, TypeScript

github.com/MuhammadUsmanGM/CodeLens

- Built a RAG pipeline over large codebases using AST-aware chunking, Qdrant vector search, and SHA-256 incremental indexing for sub-10ms retrieval across 40+ languages.
- Shipped as an npx CLI with 250+ downloads; eliminates LLM hallucinations by grounding responses in source-code truth for architectural discovery and codebase onboarding.

FerrumDB | Rust, PyO3, Tokio, AES-256-GCM, Axum

github.com/MuhammadUsmanGM/FerrumDB

- Built a custom embedded database engine in Rust (~1,000 lines) with O(1) reads/writes, AES-256-GCM encryption, atomic transactions, and a Bitcask-inspired log architecture.
- Shipped to PyPI, npm, and crates.io with 2,500+ real-world downloads across three package ecosystems; benchmarked sub-1µs GETs and 1,000 SETs in ~2-5ms via Criterion.

EDUCATION

Bachelor of Science in Software Engineering

Virtual University of Pakistan

Oct 2024 – Expected 2028

Certified Agentic AI Engineer

Presidential Initiative for Artificial Intelligence & Computing (PIAIC)

Dec 2024 – Expected 2026